



Adaptive model of thermal comfort for offices in hot and humid climates of India



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ABSTRACT

The current Indian indoor comfort standards do not reflect the country's great cultural and climatic diversity. There have been very few reports on the actual environments in Indian offices in the last three decades. We conducted a thermal comfort field study in 28 naturally ventilated (NV) and air-conditioned (AC) offices in Chennai and Hyderabad for fourteen months, and collected 6048 responses from 2787 individuals. Warm humid and composite climates are experienced in these cities, and these two climates cover about 80% area of the country.

This paper proposes an adaptive thermal comfort model for South India based on this data. Mean comfort temperature was found to be 28.0 °C in NV mode, and 26.4 °C in AC mode on all data. Chennai had slightly higher comfort temperature. We found an adaptive relationship between the prevailing outdoor temperature and the comfortable indoor temperatures. Most of the environments in NV mode and about half in AC mode were warmer than the current Indian Standard upper limit (26 °C).

In most cases, the air speed was below 0.20 m/s. Most of the subjects used fans. Air speeds of 1 m/s increased the comfort temperature by 2.7 K in both the modes. Logistic regression predicted 87% and 50% fan usage at 29 °C in NV and AC modes respectively. Several factors prevented further thermal adaptation. We can potentially improve comfort and reduce air-conditioning by providing higher air speeds with energy-efficient fans. Such strategies may be vital given the scale of the scarcity of power.

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1. Introduction

India has vast climatological diversity. The temperature standards in the country do not reflect this [1]. The National Building Code of India (NBC) recommends two temperature ranges for all the buildings in any of the five climatic zones, and the same are followed by the local city governments as well [1,2]. These were modelled using the 1992 version of ASHRAE Std-55 [3], which is now superseded. Designers in India follow Fanger's [4] predicted mean vote (PMV) model. This model relies on steady-state heat transfer theory and laboratory experiments where subjects had little to no interaction with their environments, and has many limitations when used for estimating the thermal sensation of people in real buildings [5,6].

The implications of the direct application of PMV and the uniform temperature prescriptions in real buildings, and especially in warm environments, have been debated at length across the research community [5–7]. An alternative to the PMV approach is the adaptive thermal comfort model, which is based on the idea that outdoor climate influences indoor comfort because humans can adapt to different temperatures during different times of the year. The adaptive hypothesis predicts that contextual factors, such as having access to environmental controls, and past thermal history influence building occupants' thermal expectations and preferences [8].

In practice, there has begun a paradigm shift away from the 20th century goals of cool, dry, still air, provided by isothermal environments, towards more dynamic, an isothermal environments with spatial diversity, where occupants may undertake many behavioural and physiological adaptations [9]. The old paradigm of maintaining narrow indoor temperature ranges requires enormous energy. Arens et al. [7] question the relevance of maintaining these, when there is no significant improvement found in the thermal acceptability outcomes of the building

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when compared to a building with wider temperature fluctuations. The needs of India require these kind of fundamental shifts in our thinking about how to create comfort with minimal energy use.

1.1. Thermal comfort knowledge gaps in India

There are very few thermal comfort field study reports from office buildings in India in recent years. Indraganti et al. [10] reported the comfort conditions in summer and monsoon seasons in office buildings in Chennai and Hyderabad in South India. Comfort data from India is not included in the ASHRAE 884-RP databases, which formed the basis of the adaptive model in the ASHRAE Std-55 [11,8]. Even the recent thermal comfort studies reported from the sub-continent focussed mainly on: (1) residential apartment buildings in Hyderabad [12], (2) vernacular housing in the north-east region [13], (3) a railway terminal [14], (4) traditional housing in Kerala [15], (5) hostel buildings [16], (6) post occupancy evaluation of an office building [17] and (6) post occupancy evaluation for energy efficient design of Indian and UK offices [18]. Sharma and Ali's [19] research focussed on naturally ventilated buildings in Roorkie in North India. They recorded the comfort responses of 18 office occupants (six in each year) over a period of three consecutive summer seasons in circa 1978.

Office occupants and their comfort conditions in India are quite different from their counterparts from the west. These need to be investigated in detail in order to develop custom made comfort standards, which is a long drawn process. Field studies of actual thermal conditions and occupant response would also provide vital guidance for both the design and operation of new buildings or retrofits.

1.2. Power shortages in India

The energy deficit in India is increasing, with many associated economic implications [20]. The power shortages ranged from 3 to 18% all across the nation in 2013, with the southern region facing the highest shortage. The shortages increased significantly in the last few years [21]. For example, power shortages increased from 4.1 to 18.9% in Andhra Pradesh and 2.8–23.6% in Tamil Nadu over the last five years. The anticipated power deficit for the year 2012–13 in these two South Indian states of Andhra Pradesh and Tamil Nadu is 22.8% and 29.6% respectively. Simultaneously, the power consumption in the last half-decade has gone up by about 13.6% from 2004 to 2010 [21–23], with building sector consuming 187.9 million tons of oil equivalent [23]. This sector has the greatest energy saving potential, and the application of an adaptive model of thermal comfort based on field studies is a practical first step in achieving sustainable thermal comfort indoors [24].

1.3. Objectives

With the aforementioned issues in mind, this paper has the following objectives:

1. To investigate the thermal comfort conditions in Indian buildings used by the regular occupants
2. To study in detail the adaptation of the office occupants and to evaluate the actual comfort temperatures based on the field study data
3. To develop an adaptive model for Indian offices in these climatic zones

4. To find field evidence for the contribution of fans in achieving comfort in warm environments of India.

We undertook thermal comfort field studies for 14 months in 28 offices in two south Indian states, Andhra Pradesh and Tamil Nadu, during January 2012 to February 2013. We investigated the present thermal comfort conditions in these offices and examined the adaptive methods in use, aiming towards the development of an adaptive model of thermal comfort for this region. Indraganti et al. [10] reported the preliminary findings from the summer and southwest monsoon seasons, while this paper analyses the comfort conditions observed throughout the year.

2. Field survey

The field surveys were conducted in Chennai and Hyderabad, the capital cities of Tamil Nadu and Andhra Pradesh states in South India. Chennai (N13°04' and E80° 17' and 6.7 m above the mean sea level) has a warm, humid, wetland coastal climate. Hyderabad (N17°27' and E78° 28' and 540 m above the mean sea level) has a composite climate.

About 80% of the geographical area of the country experiences climates similar to these two cities. Both the climates have four seasons: winter, summer, southwest monsoon (SWM) and north-east monsoon (NEM). Overall, we collected 6048 sets of data in fourteen months of survey (Table 1).

2.1. The buildings surveyed

Twenty-eight ordinary office buildings located within a radius of 12–13 km of a metropolitan area were selected for the field study. The selection criteria included that the building should be (1) well-maintained, (2) under a management with occupants who would allow and participate in our year-long non-invasive survey, and (3) an office building in typical construction (brick/concrete and glass), where there are more than ten people working. There were originally 26 buildings, labelled C1 – C13 in Chennai and H1 – H13 in Hyderabad. Offices functioning in C5 and H11 were shifted to two new buildings, labelled as C5a and H11a respectively, and the survey was continued at these new locations, thus bringing the total number of buildings to 28.

In Chennai most of the surveyed buildings functioned in air-conditioned (AC) mode and in Hyderabad most of them had both (i.e. they combined operable windows with mechanical cooling, and switched between NV and AC modes.) Due to the state -wide power deficit [21], all the buildings faced a daily blackout for at least two hours throughout the year in Chennai, and in summer and SWM seasons in Hyderabad [25]. Buildings C3 and H7 had diesel operated on-site power generators to run

Table 1
Details of the data collection.

Case	Season (months)	NV (N)	AC (N)	AC _{off} (N)
Chennai	Winter (1 and 2)	38	446	81
	Summer (3, 4 and 5)	30	605	137
	SWM (6, 7, 8 and 9)	7	617	33
	NEM (10, 11 and 12)	57	854	2
Hyderabad	Winter (12, 1 and 2)	834	779	66
	Summer (3, 4 and 5)	112	563	40
	SWM (6, 7, 8 and 9)	177	268	23
	NEM (10 and 11)	97	178	4
All	1 to 12	1352	4310	386

N: Number of datasets collected; SWM: Southwest monsoon; NEM: Northeast monsoon.

the AC units in addition to the lights and computers between 9 am and 5 pm, during the outages. However, due to the operation and maintenance difficulties, such generators could not sometimes cater to the AC units during the power breakdowns. Therefore, the data collected during an outage in an AC building with all the AC units switched off, is noted down as 'AC_{off}' mode. All the other offices had smaller generators to power the computers, lights and in some cases ceiling fans during a power-cut. Sometimes longer power cuts for a few more hours in a day were imposed due to further power shortages. This paper primarily focuses on NV and AC modes.

2.2. Field investigation

This is a long-term survey. We repeated the survey after every 3–6 weeks in all the buildings, between 9 am and 11 pm on working days. A few office workers customarily worked late in the evening and continued to participate in the survey during those late hours. However, as the survey progressed, some buildings refused access and a few participants stopped responding. Hence the monthly sample size varied slightly. The surveys lasted approximately 7–12 days in a month. The first author personally conducted all the surveys once every season, and in the rest of the months two trained local surveyors conducted the surveys. Each office was visited once or twice a month. The methods are further elaborated in Indraganti et al. [10]. Calibrated digital instruments were used and are described below.

2.3. Questionnaires and scales

We prepared the questionnaires in English first and then translated them into Telugu and Tamil, the two languages spoken locally. The English version was based on McCartney and Nicol [26] and the Telugu version was modified from Indraganti [12].

The questionnaire had three parts: personal identifiers, thermal responses and preferences, and behavioural adaptation. Each respondent filled the questionnaire once or sometimes twice a month.

Thermal sensation was measured using ASHRAE's seven-point scale ranging from *hot* (+3) to *cold* (−3), with *neutral* (0) in the centre. Nicol's five-point scale measured the thermal preference. A binary input of 0 = acceptable or 1 = unacceptable indicated the thermal acceptability, as shown in Table 2.

2.4. Outdoor and indoor environmental data and personal variables

Outdoor environmental data consisted of daily maximum and minimum temperature and relative humidity values for all the fourteen months of survey, collected from the Indian

meteorological data stations located at the local airports [27]. Logistic and financial constraints prevented us from collecting this data in-situ.

Calibrated quick-response digital instruments were used to measure the indoor environment (Table 3). The measurements were recorded after the instruments and their readings had stabilized in the surveyed area. Although the instruments each had different time-constants, it took approximately 5–7 min for all of them to stabilize. Anemometers were calibrated using the low velocity wind tunnel at The University of Tokyo. The thermometers were procured pre-calibrated and were certified.

The instruments were fixed to a small folding platform, which was placed on the subjects' table and measured 1.1 m from the floor, as shown in Fig. 1. The surveyors noted down the environmental measurements, clothing and metabolic activity of the respondents while they were filling the questionnaires, along with the personal and environmental controls in use during the survey. The environmental controls observed were: windows, doors, curtains/blinds, ceiling and wall-mounted fans, lights, air-conditioners, evaporative coolers (locally termed as air coolers), table lights, pedestal fans, hand fans and personal table fans.

3. The subject sample

3.1. The subject sample size, age and gender

The subjects were all Indian nationals in the age group of 18–68 years who lived in the respective city and had worked in the offices for longer than three months. A total of 2787 subjects responded: 2157 males and 630 females.

The average age of the subjects was 31.4 yrs (standard deviation (SD) = 9.5 yrs and sample size (N) = 2686). Eighty-two men and nineteen women refused to provide their age during the survey. The age and gender profile of the subject sample in both the cities is similar as shown in Fig. 2.

Clothing and metabolic rate were estimated using tables from ASHRAE Standard - 55 [11]. Clothing insulation of subjects was obtained using the summation formula $I_{cl,tot} = \sum I_{cl,i}$ where, $I_{cl,tot}$ was the insulation of the entire ensemble and $I_{cl,i}$ was the insulation of an individual piece of garment. We added 0.04 clo for all the subjects to account for the insulation of the undergarments assumed to be worn, an upholstery insulation of 0.15 clo to all the seated subjects, and 0.15 clo to the subjects wearing a sari, to account for the added insulation of a petticoat, which is most often worn under a sari.

Typical clothing ensembles used by the subjects are shown in Fig. 1. The clothing insulation of saris was estimated using an equation provided by Hanada et al. [28,29] to estimate clo value from the weight of an ensemble, and the insulation of *chudidhar-kameez* is estimated similar to Indraganti [12]. The methods and

Table 2
Thermal comfort scales used.

Scale value	Description of scale		
	Thermal sensation (TS)	Thermal preference (TP)	Thermal acceptability (TA)
3	Hot		
2	Warm	Much cooler	
1	Slightly warm	A bit cooler	Unacceptable
0	Neutral	No change	Acceptable
−1	Slightly cool	A bit warmer	
−2	Cool	Much warmer	
−3	Cold		

Table 3
Details of the instruments used for the environmental measurement.

Description	Trade name	Parameter measured	Range	Accuracy
Thermo-hygro-CO ₂ meter	TR-76Ui	Air temperature	0 to 45 °C	±0.5 °C
		Humidity	10 to 90 %RH	±5%
		CO ₂ level	0 to 5000 ppm	±50 ppm
Probe thermometer with black painted table tennis ball	Tr-52i	Globe temperature	(−60 to 155 °C)	±0.5 °C
Hot wire anemometer	Testo −405	Air velocity	0.01–50.0 m/s	±0.01 m/s



Fig. 1. (1–5) Survey environments in (1) Chennai – AC, (2) Chennai – NV, (3) Hyderabad – AC, (4) Chennai – AC and (5) Hyderabad – NV; (6) The instrument set up; (7–9) Typical clothing ensembles of the subjects; (S): Woolen shawls at the subject's immediate disposal in AC offices in Hyderabad; (A) Thermo-hygro-CO₂ meter; (B) Hot-wire Anemometer; and (C) Globe thermometer.

the subject sample were presented in a greater detail earlier in Indraganti et al. [10].

4. Results and discussion

4.1. Physical environmental measurements

4.1.1. Outdoor environments and the running mean temperature

During the survey period, the outdoor daily mean temperature (T_m) in Chennai varied from 24.5 to 35.5 °C (SD = 2.8 °C), while daily mean relative humidity varied from 40 to 95%. Hyderabad recorded slightly lower temperature, which varied from 21.5 to 34.5 °C (SD = 3.7 °C), with a wider range of daily mean relative humidity, ranging between 17 and 91% (Table 4, Fig. 3).

Hot-humid summer conditions are experienced in Chennai from March to May, often extending into June by a few weeks. Hyderabad summer is characterized by predominantly hot-dry weather. This season has the highest levels of discomfort in both the cities. Monsoon seasons are comparatively cooler. Winters during the survey period were short and were very mild.

Researchers have made use of the running mean temperature as it reflects the people's responses to the outdoor environment better

than the outdoor daily mean temperature [30]. The exponentially weighted running means give greatest weight to temperatures in the most recent past, and progressively less weight to those in the more remote past.

The exponentially weighted running mean of the daily mean outdoor temperature is obtained from the mathematical expression:

$$T_{rm}(\text{tomorrow}) = (\alpha)T_{rm}(\text{yesterday}) + (1 - \alpha)T_m(\text{today}) \quad (1)$$

Where, T_{rm} is the running mean temperature (°C), T_m is the outdoor daily mean temperature (°C), and α is a unitless constant between 0 and 1 and is usually taken as 0.8. It indicates a half-life of approximately 3.5 days [30]. We estimated T_{rm} for all the days from January 2012 to February 2013. As can be observed from Table 4 and Fig. 3, Chennai is relatively warmer and more humid than Hyderabad most of the year.

4.1.2. Hygro-thermal conditions: indoors

Indoor conditions in Chennai were also slightly warmer and more humid than those of Hyderabad (Table 4, Fig. 4). The difference is statistically significant at 95% Confidence Interval (CI) in both NV and AC modes. Similarly NV environments were significantly warmer than AC at 95% CI.

The NV mode experienced a higher mean indoor globe temperature (T_g mean = 28.8 °C) than AC mode (T_g mean = 26.2 °C) during the survey. In about 8% of cases (all data) relative humidity was outside of the 30–65% levels. It is important to note that only in 37.4% cases (on all data) the indoor temperature was within the comfort band of 21–26 °C, specified in the Indian codes [1].

The indoor and outdoor variables had a normal distribution [31]. Pearson's correlation between various indoor and outdoor variables was estimated (Table 5). As expected, the indoor temperatures followed the outdoor temperatures more closely in NV mode in both the cities than in AC mode.

4.1.3. Indoor air movement

Subjects working in NV mode experienced higher indoor air speeds than in AC mode, apparently through cross ventilation and through the use of fans and in a few cases evaporative coolers. Chennai recorded significantly higher indoor air speeds than Hyderabad in all the modes at 95% CI (Table 4, Fig. 4b).

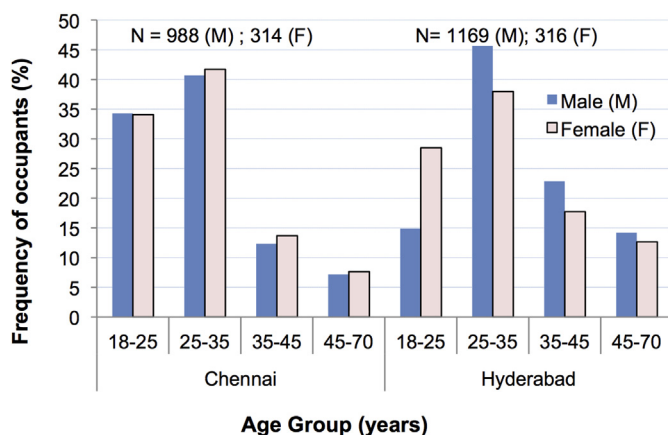


Fig. 2. Age and gender distribution of the respondents during the fourteen months of the survey (N = Number of occupants).

Table 4

Indoor and outdoor environmental variables observed.

Variable	Chennai				Hyderabad				All			
	NV		AC		NV		AC		NV		AC	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Sample size	132		2522		1220		1788		1352		4310	
T_m (°C)	26.9	2.7	28.9	2.8	25.4	3.0	27.5	3.9	25.5	3.0	28.4	3.4
T_{rm} (°C)	27.1	2.3	28.9	2.6	25.5	2.4	27.3	3.3	25.6	2.4	28.2	3.0
T_a (°C)	29.7	2.2	26.5	1.5	29.0	1.9	26.1	1.6	29.1	1.9	26.3	1.6
T_g (°C)	29.5	2.1	26.7	1.4	28.7	2.0	25.6	1.7	28.8	2.0	26.2	1.6
RH (%)	60	6	50	8	43	11	46	11	45	12	48	9
W_v (g _w /kg _{da})	15.7	2.7	10.8	2.0	10.7	2.7	9.5	2.1	11.2	3.1	10.3	2.1
V_a (m/s)	0.50	0.30	0.20	0.20	0.10	0.20	0.00	0.10	0.17	0.25	0.11	0.17
CO ₂ (ppm)	821	622	1370	459	860	291	962	311	856	337	1200	452
I_{cl_tot} (clo)	0.69	0.08	0.71	0.09	0.71	0.08	0.69	0.06	0.70	0.08	0.70	0.08
Activity (Met)	1.0	0.1	1.0	0.1	1.0	0.1	1.0	0.0	1.0	0.1	1.0	0.1

T_m : Outdoor daily mean temperature (°C), T_{rm} : Outdoor daily running mean temperature (°C), T_a : Indoor air temperature (°C), T_g : Indoor globe temperature (°C), RH: Indoor relative humidity (%), W_v : Indoor absolute humidity (g_w/kg_{da}), V_a : Indoor air velocity (m/s), CO₂: CO₂ concentration indoors (ppm), I_{cl_tot} : Total clothing insulation (clo).

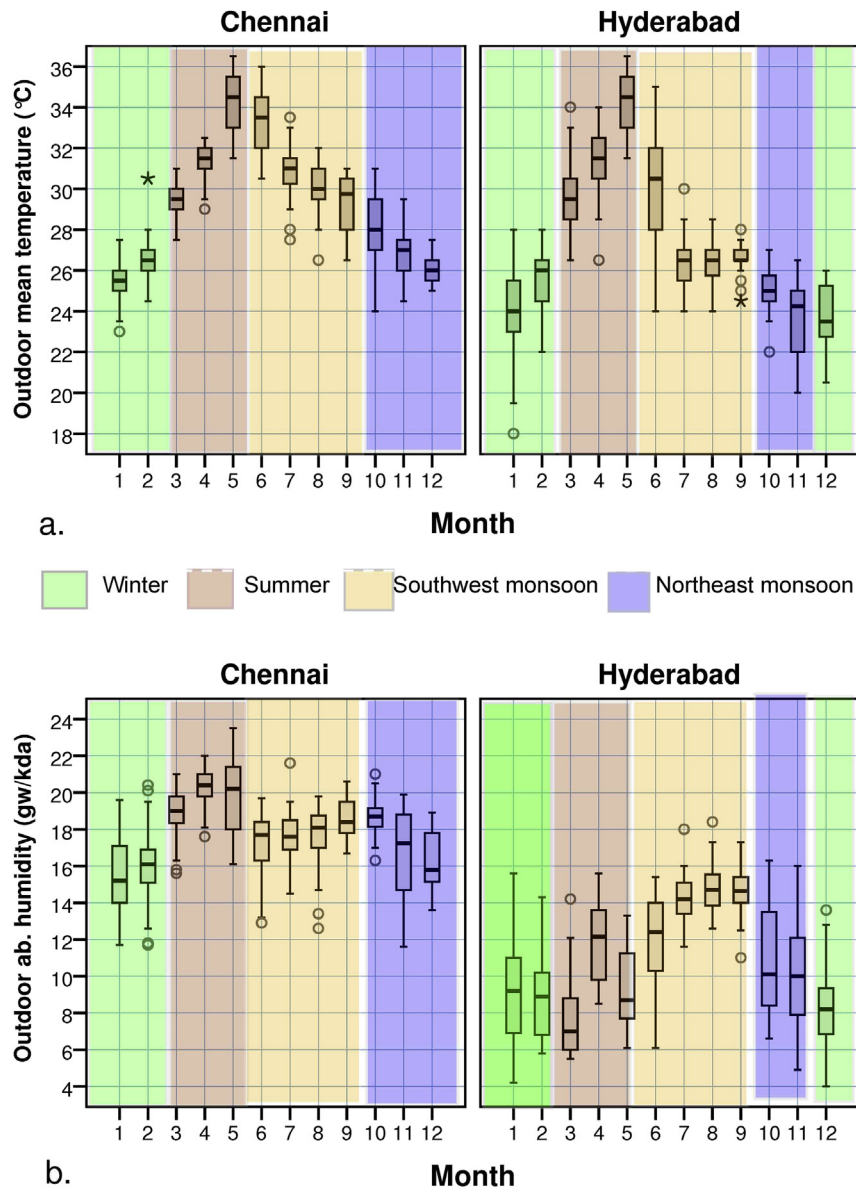


Fig. 3. Box-plot indicating the (a) Variation in the outdoor daily mean temperature (T_m) (°C) and (b) Outdoor daily mean absolute humidity (W_{vout}) (g_w/kda) during the period of survey (January 2012 to February 2013) in Chennai and Hyderabad cities. Chennai is warmer and more humid than Hyderabad most of the year. Also shown are the seasons.

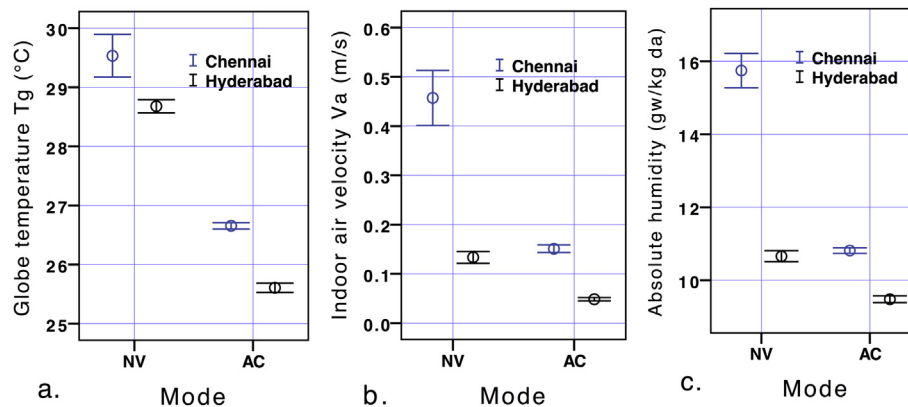


Fig. 4. Intercity and modal variation in (a) Globe temperature, (b) Air movement and (c) Absolute humidity observed indoors between Chennai and Hyderabad, (All data). Chennai was significantly humid and recorded higher indoor temperatures and air velocities. Error bars indicate 95% confidence interval.

ASHRAE Std-55-2010 [11] suggests that indoor operative temperature can be elevated above the still air limits up to a maximum of 3 K, if there are also higher air speeds ranging from 0.2 to 0.8 m/s for lightly clothed (0.5–0.7 clo) occupants in sedentary activities. This requires that occupants have control of the air speed. In comparison, in 80% of the environments in NV mode in Chennai, the measured air velocity was less than 0.60 m/s, compared to 0.19 m/s in AC mode. Air movement in AC mode was due to the operation of ceiling and/or wall mounted/pedestal fans and in some cases both. Hyderabad had even lower air velocities in comparison, with V_a less than 0.20 m/s in 80% of the cases in NV mode. Interestingly, near still air conditions prevailed in AC mode, with less than 0.05 m/s air speed observed in 80% of the cases.

There is ample evidence in literature supporting the idea that India can take advantage of higher air speeds than those observed, in order to offset discomfort at elevated temperatures. For example, Nicol and Roaf [32] observed occupants accepting air speeds up to 1.5 m/s in their Pakistan studies. Cheng and Ng [33], also noted air speeds from 1.0 to 1.5 m/s, satisfying 80% of the occupants in Hong Kong in naturally ventilated offices. Zhai et al. [34] report from a climate chamber study of achieving occupant thermal comfort at 30 °C and 60% relative humidity with floor mounted personally controlled air movement. Adding further, they state that, ‘relative perceived air quality’ was maintained up to 30 °C and 80% RH without discomforting the subject from humidity, air movement or eye dryness.

4.2. Personal variables: activity and clothing insulation

We observed the metabolic rate varying from 0.7 Met (resting) to 2.0 Met (standing active work) during the survey. We found a few

Table 5
Strength of the relationship between the environmental variables.

Variable	Correlation coefficient with outdoor daily mean temperature (T_m) ^a			
	Chennai		Hyderabad	
	NV	AC	NV	AC
T_{rm} (°C)	1.0	0.9	0.9	1.0
T_a (°C)	0.9	0.1	0.7	0.3
T_g (°C)	0.9	0.2	0.7	0.3
W_v (g _w /kg _{da})	0.8	−0.1	0.1	−0.5
V_a (m/s)	0.5	0.1	0.2	0.4

^a Correlation coefficient is significant at 0.01 level (2 tailed); T_g : Indoor globe temperature (°C); T_a : Indoor air temperature (°C); W_v : Indoor absolute humidity (g_w/kg_{da}), and V_a : Indoor air velocity (m/s).

subjects at rest, perhaps taking short post-meal siestas, or sometimes during the outages, or when the indoor environment was warmer than the usual. However, the dependency of metabolic rate on the indoor globe temperature (T_g) was not statistically significant in Chennai ($p > 0.05$) while the same was significant in Hyderabad ($p < 0.001$).

Total clothing insulation (I_{cl_tot}) varied slightly during the survey (mean = 0.70 clo, min = 0.47, max = 1.2, SD = 0.08 clo, $N = 6048$). This included the upholstery insulation of 0.15 clo added to all the seated subjects. The total clothing insulation of subject’s ensemble (without the chair insulation) also varied similarly (mean = 0.55 clo, min = 0.38, max = 1.05, SD = 0.08 clo, $N = 6048$). The mean total clothing insulation varied from 0.68 clo in summer to 0.72 clo in monsoon and winter seasons. Most men wore formal trousers or jeans (99%) and half of them were barefooted during our survey. About 66% of male subjects were dressed in full sleeve shirts as most offices imposed an office dress code.

Women’s total clothing insulation (mean = 0.78 clo; SD = 0.12 clo) was a bit heavier and varied slightly more than men’s (mean = 0.68 clo; SD = 0.04 clo). Some assumptions had to be made about the clo value of particular articles of Indian style clothing where the information did not exist in existing tables. Most of the female subjects (93%) were dressed in Indian ensembles. These were sari and chudidhar/salwar–kameez (CSK) with or without a dupatta (thin cotton or polyester/silk shawl-like accoutrement), as shown in Fig. 1: 8 and 9. The CSK was the most preferred attire for women in Indian offices, used by 71% of the female subjects. A few women wore small headscarves catering to their religious needs.

We determined the clothing insulation of a sari with blouse from the weight the ensembles [28,29]. On the other hand, the clo value of pyjamas from ASHRAE [11] was taken as a near equivalent for the CSK. In AC offices in Hyderabad, we observed women maintaining woollen shawls or sweaters at their immediate disposal even in summer (Fig. 1: 3). These were used during cold drafts. Some men abstained from using footwear for religious obeisance usually during the winter (Fig. 1: 7).

4.3. Subjective thermal variables

4.3.1. Thermal sensation (TS)

We used the question “how do you feel the present temperature of this room?” to elicit the occupants’ subjective thermal sensation. The subjects chose from the ASHRAE’s seven point scale of sensation, which varied from hot (3) to cold (−3) with the neutral sensation at the centre (0) (Table 2).

Although the wording of the question is slightly different than what is traditionally associated with the thermal sensation scale, changes were made based on past experiences with Indian subjects' misinterpretation of such wording. Since the questionnaire was administered personally rather than online, we were also able to clarify the question verbally to them to ensure they expressed their own current feelings about the immediate indoor environment, and not give us a number about the room temperature, or talk about past experiences.

The mean value of the sensation vote was higher than neutral in NV mode and lower in AC mode (Fig. 5 and Table 6), and these differences are statistically significant. The survey was done in warm climates and it is imperative to note that Chennai also experienced high humidity. Therefore, on the whole, a higher percentage of subjects had 'warm and hot' sensations than 'cool and cold' sensations.

Fanger's PMV over-estimated the actual sensation as seen in Table 6 in both the cities in all the modes, and the differences are

statistically significant (95% CI). This is consistent with what has been found by other researchers [5,12,35–37].

4.3.2. Thermal preference (TP)

A preference for cooler environments is very evident throughout the year (Fig. 5 and Table 6). The mean preference vote (TP_{mean}) in NV mode (0.4) was slightly higher than in AC mode (0.3) and the difference is statistically significant in Hyderabad. We did not observe any significant inter-city differences in the mean preference vote.

4.3.3. Thermal acceptability (TA) and predicted percentage of dissatisfied (PPD)

Many physio-psychological variables influence thermal acceptability, which is therefore a very complex construct [38]. In practice, however, thermal acceptability is often assessed with a very simple scale. In this study, thermal acceptability was measured as a binary variable (0: acceptable, 1: unacceptable) from the responses to a

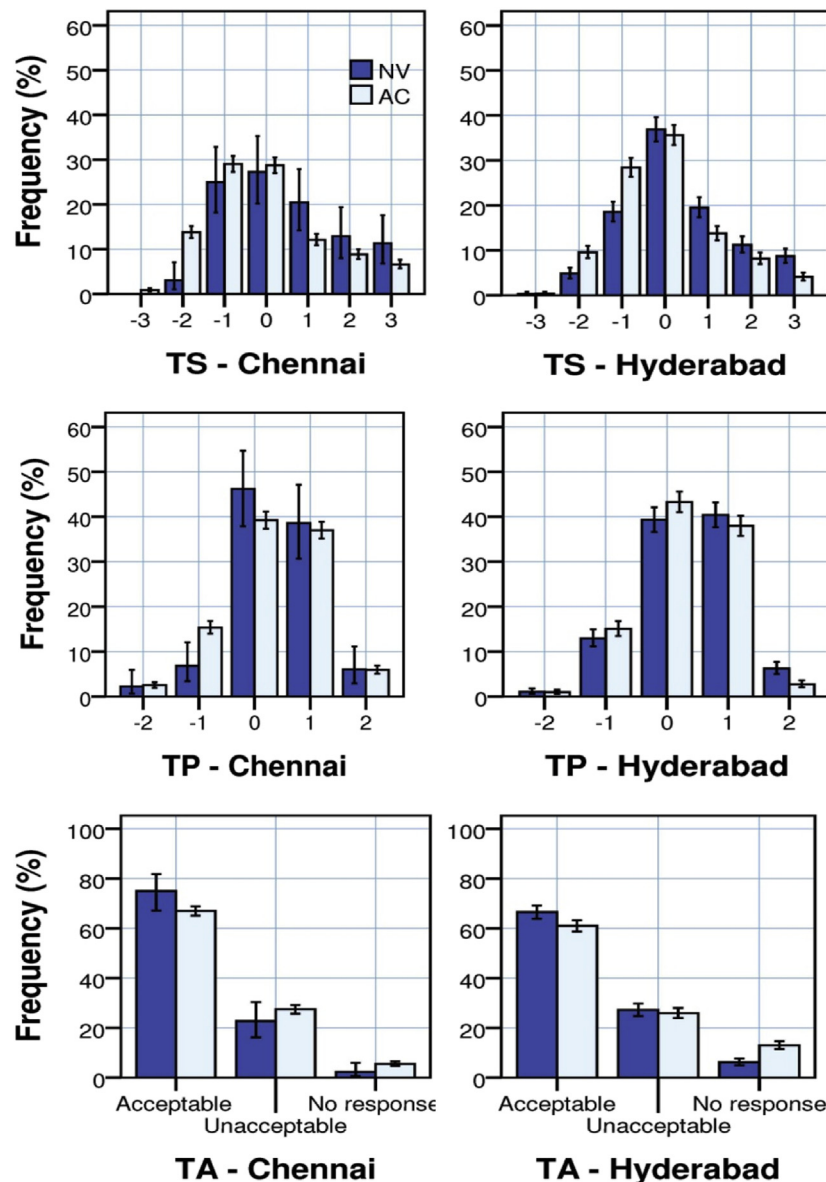


Fig. 5. Distribution of subjective thermal responses in Chennai and Hyderabad (all data). A majority voted on the warmer side of the sensation scale, preferred cooler temperatures and the acceptability outcome did not vary significantly with the mode of operation in both the cities. The error bars represent 95% confidence interval.

Table 6
Descriptive statistics of measured subjective thermal comfort variables.

Variable	Chennai				Hyderabad				All			
	NV		AC		NV		AC		NV		AC	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Sample size	132		2522		1220		1788		1352		4310	
TS	0.5	1.4	−0.1	1.4	0.4	1.3	−0.1	1.2	0.4	1.3	−0.1	1.3
PMV	1.4	0.9	0.5	0.5	1.2	0.6	0.3	0.5	1.2	0.6	0.4	0.5
TP	0.4	0.8	0.3	0.9	0.4	0.8	0.3	0.8	0.4	0.8	0.3	0.8
TA_n	0.23	0.42	0.29	0.45	0.29	0.45	0.30	0.46	0.28	0.45	0.29	0.46
PPD (%)	48.8	31.8	15.9	13.1	37.9	23.6	11.7	11.4	39.0	24.7	14.2	12.6

TS: Thermal Sensation; PMV: Predicted Mean Vote; TP: Thermal Preference; TA_n: Thermal non-acceptability; PPD: Predicted Percentage Dissatisfied.

direct question, ‘do you accept the present indoor conditions?’ (Table 6, Fig. 5).

About 28–29% of the subjects voted the environments unacceptable in both NV and AC modes while 7.9% of the subjects abstained from giving their TA vote ($N = 476$). The differences in thermal acceptability outcomes did not vary much with the mode of operation in these two cities, nor were there any inter-city differences as can be seen in Fig. 5 and Table 6. Lower thermal acceptability was perhaps partly due to frequent and daily power outages.

Thermal dissatisfaction as estimated through Predicted Percentage Dissatisfied (PPD) was much higher in NV mode (39%). PPD is derived from PMV, and is based on the assumption that thermal sensation votes of $\pm 2, 3$ represent ‘dissatisfied’. PMV was observed to be over estimating the actual sensation, more so in NV mode. This in part explains why PPD was also much higher in NV mode.

The findings of Arens et al. [7] are important to note here. They found that the acceptability outcomes of tightly air temperature controlled buildings are no different from the non-tightly controlled environments, based on analysis of comfort data from Berkeley City Centre project [39] and the ASHRAE database [11,8]. Arens et al. [7] further noted from the European SCATs database [26] that the occupant’s overall assessment of the indoor environment is not affected very much by the annual temperature fluctuations that take place in a building.

4.4. The comfort temperature

4.4.1. Comfort temperature by linear regression

Linear regression of thermal sensation and indoor globe temperature is one of the methods for evaluating the neutral temperature in field studies. Using the regression, neutral temperature is determined by setting the thermal sensation to neutral (zero). Our linear regression for NV buildings is:

NV mode:

$$TS = 0.26T_g - 7.09 \quad (2)$$

($N = 1352$, $r = 0.40$, standard error of the regression coefficient (S.E.) = 0.016, $p < 0.001$).

This is presented in Fig. 6 along with lines indicating 95% confidence intervals (CI) of slope of the equation.

The regression for AC buildings was not likely to be meaningful. The data used in the linear regression is pooled from 28 buildings through fourteen months in NV and AC modes. In order to eliminate the possibility of the ‘between-buildings’ effect being significant, we subjected the data to the univariate general linear regression. In this procedure, the building code is used as a fixed factor and indoor globe temperature as the covariate. We found no significant inter-building differences in NV mode, but in AC mode we found significant differences. This could be due to some irregularity or

roughness of data from within the buildings. This may reduce the possibility of obtaining a meaningful regression of sensation with indoor temperature in AC mode.

The slope of the Eq. (2) is comparable to others’ results. Karyono [40] noted a slope of 0.32 K^{-1} in Thai offices, Indraganti [12] recorded 0.31 K^{-1} in the residential environments in Hyderabad, while Nicol and Roaf [32] observed 0.15 K^{-1} in Pakistan, the latter indicating a much higher adaptation in Pakistani subjects.

Using the Eq. (2) we estimated the regression neutral temperature (T_n) for NV mode. On all data, it yielded the neutral temperature as $27.3 \text{ }^\circ\text{C}$. As the regression method failed to produce a reliable model for the neutral temperature in AC mode, we considered using the Griffith’s method [41]. Other researchers have also used this method for predicting the comfort temperature from the field survey data [10,26,30,37].

4.5. Griffith’s comfort temperature (T_{conf})

In this method, a ‘comfort temperature’ (T_{conf}) is derived from every dataset using an assumed slope (for example, 0.5 K^{-1}). It is the indoor globe temperature at which the respondent would record thermal neutrality if the divisions on the ASHRAE scale correspond to a change in indoor globe temperature (of 2 K). Humphreys et al. [30] note that each ‘spot-value’ of the comfort temperature is based on a single thermal sensation on a single occasion and therefore subject to considerable random uncertainty, but not to systematic error if the chosen value of 2 K is correct.

We calculated the Griffith’s temperature for all votes using 0.50 K^{-1} as Griffith’s slope, which has support from other experiments [30,37,42]. With this method we obtained a mean comfort

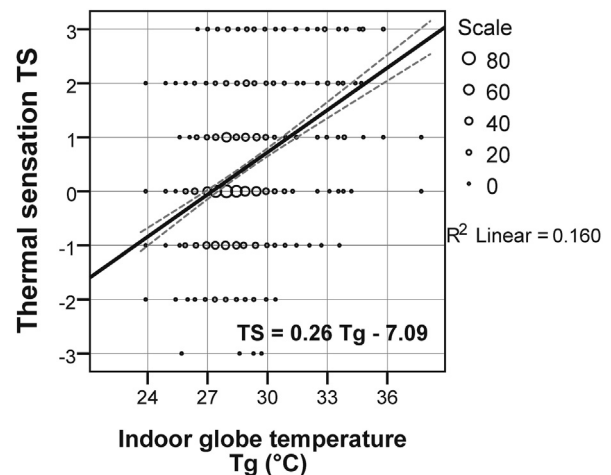


Fig. 6. Change in thermal sensation with indoor globe temperature for NV mode, with dashed lines indicating 95% CI of slope.

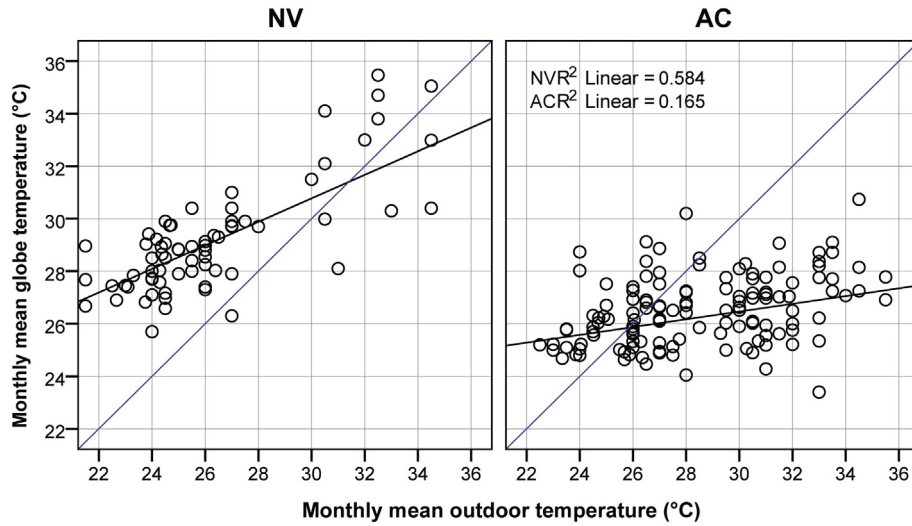


Fig. 7. Change in monthly mean indoor globe temperature with monthly mean outdoor temperature for both NV and AC modes. Each point represents the aggregated data for a single building in a month.

temperature of 28.0 °C in NV mode and 26.4 °C in AC mode on all data. This finding is in close agreement with the neutral operative temperature observed in Singapore: 28.5 °C in NV mode and 24.2 °C in AC mode [43].

In order to fully comprehend the dependence of the indoor temperature on the outdoor temperature, we aggregated the monthly outdoor and indoor globe temperatures in all the buildings separately for both the modes. The aggregated monthly mean outdoor temperature is then regressed against the monthly mean indoor globe temperature as shown in Fig. 7. We obtained the regression coefficient of 0.45 ($r = 0.76$, S.E. = 0.010, $p < 0.001$) for NV mode, and 0.15 ($r = 0.41$, S.E. = 0.005, $p < 0.001$) for AC mode (Fig. 7).

Interestingly, the regression coefficient of 0.45 obtained for NV mode matched very closely with that of 0.50 obtained by Humphreys [44]. It means that for every degree rise in outdoor temperature, the indoor temperature increases only by about half a degree. The controls like windows and blinds are partly successful in modulating the indoor temperatures.

In contrast, the AC buildings are much less dependent on the outdoor temperature, as would be expected because they are using energy to cool the buildings, so as to keep a fairly constant indoor temperature.

4.6. Adaptive model of thermal comfort

A consequence of the adaptive theory of thermal comfort is that the comfortable temperatures indoors can be related to the climate. We regressed indoor comfort temperature against outdoor running mean temperature and obtained the following adaptive model.

NV mode:

$$T_{\text{comf}} = 0.26T_{\text{rm}} + 21.4 \quad (3)$$

($N = 1352$, $r = 0.24$, S.E. = 0.028, $p < 0.001$, F statistic = 82.5).

AC mode:

$$T_{\text{comf}} = 0.15T_{\text{rm}} + 22.1 \quad (4)$$

($N = 4310$, $r = 0.16$, S.E. = 0.014, $p < 0.001$, F statistic = 115.9).

The adaptive model of thermal comfort for the warm and humid climates of South India is presented in Fig. 8. It shows the dependency of indoor comfort temperature on the daily outdoor running mean temperature for NV mode (Eq. (3)). A similar relationship is also shown along side for AC mode (Eq. (4)). These two equations are statistically significant at $p < 0.001$ and their slopes are significantly different from each other at 95% CI.

4.7. A comparison with NBC, CEN, CIBSE and ASHRAE standards

We compared the Indian data with the CEN (Comité Européen de Normalisation, or European Committee for Standardization) standard for NV mode (Fig. 9a) and with the CIBSE (The Chartered Institution of Building Services Engineers) Guide [45] for the AC

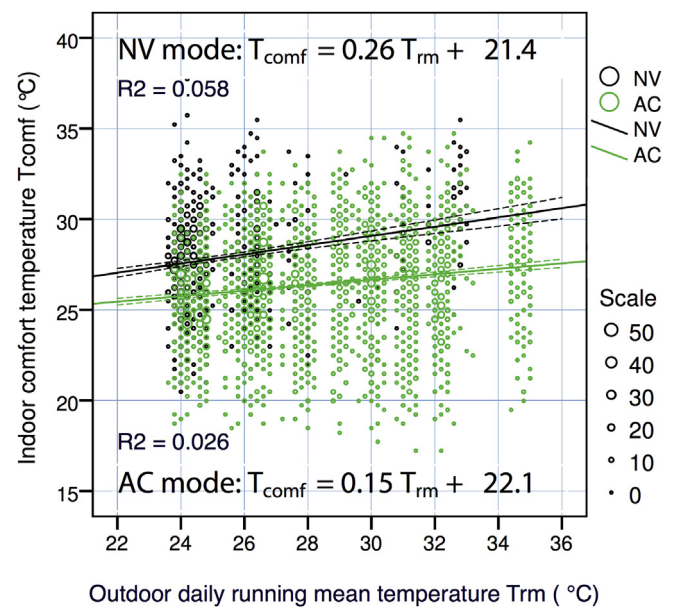


Fig. 8. Adaptive model of thermal comfort for warm humid climates of south India for NV mode. Also featuring is the similar relationship observed for AC mode, at $p < 0.001$. Dotted lines indicate 95% CI of slope.

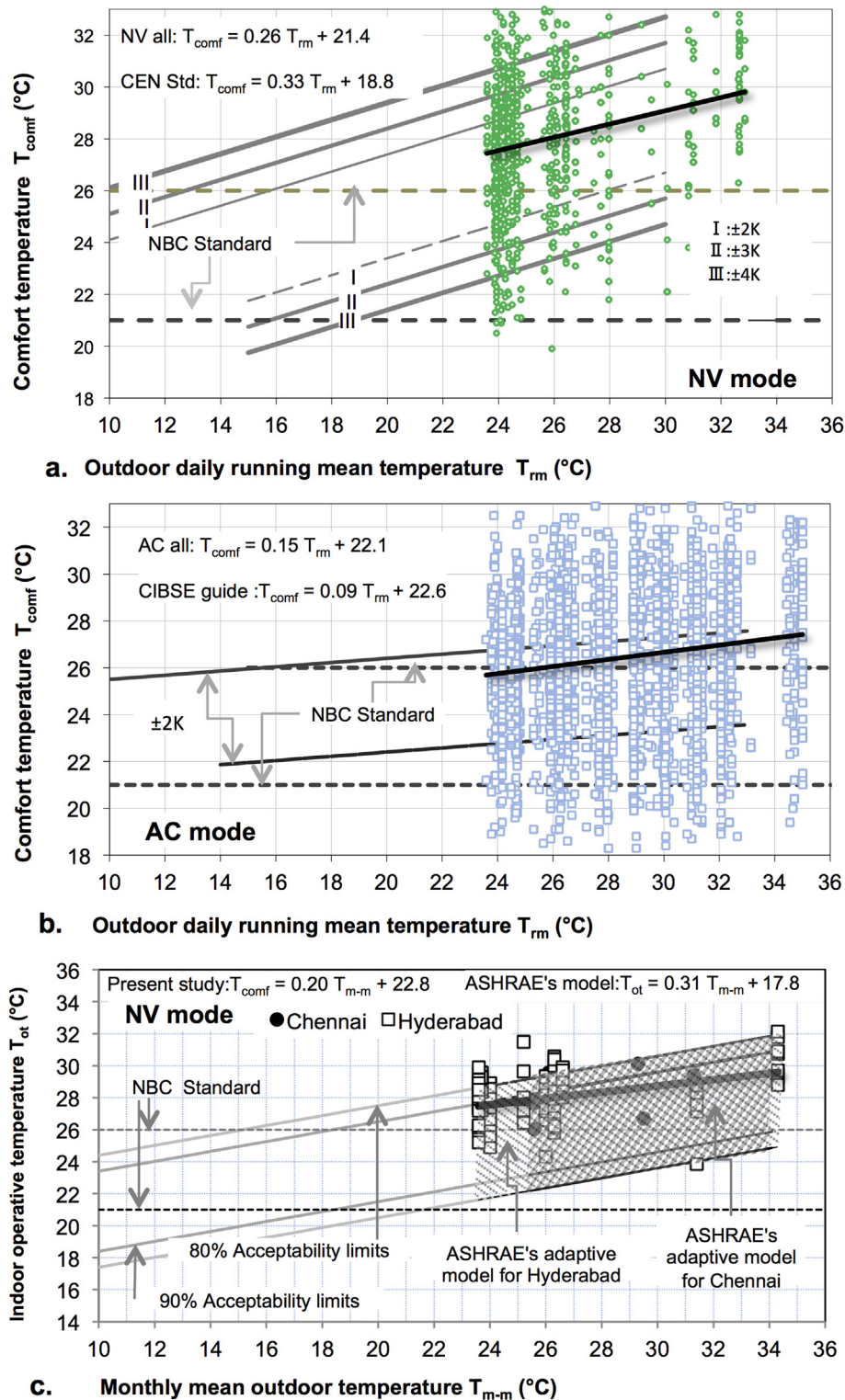


Fig. 9. A comparison with NBC, CEN, CIBSE and ASHRAE standards; (a) Adaptive model of thermal comfort for NV mode and CEN standard; (b) Comfort data of AC mode superimposed over the relationship mentioned in the CIBSE guide. Each point represents a single response; (c) ASHRAE's adaptive model of thermal comfort for Chennai (double hatched area) and Hyderabad (hatched area) with NBC standard (dotted lines). Also shown superimposed are the comfort data from the present survey. Each point represents the aggregated comfort temperature data for each floor in all the buildings over a month taken as the unit of analysis for NV mode.

mode (Fig. 9b). The present data is also super-imposed on the ASHRAE's adaptive model (Fig. 9c). We also used the Indian NBC (National Building Code) standard for making a comparison with the Indian data.

4.7.1. A comparison with NBC and the European data

Importantly, in NV mode, almost all the data points and the regression line lie above the NBC standard. This clearly indicates that the NBC standard is too conservative and people are

comfortable at warmer temperatures. In AC mode, while most of the points are above the upper limit of the NBC, the regression line too crosses the upper boundary. This confirms the need for its revision as mentioned earlier by Indraganti [12].

For NV mode, we obtained the slope of the regression equation as 0.26 as compared to 0.33, the slope of the adaptive model for free running buildings in CEN standard [46]. Indoor comfort temperature in Indian offices in NV mode changed more gradually with the outdoor temperature than it did in Europe (Fig. 9a). Conversely, in AC mode, when compared to the CIBSE guide (slope: 0.09), the comfort temperature varied relatively steeply with the outdoor temperature (slope: 0.15). The subjects' comfort temperatures were higher in NV mode (mean = 28.0 °C, SD = 2.6) than in AC mode (mean = 26.4 °C, SD = 2.8), similar to the finding of others [47,43,48]. In comparison with the European subjects, the Indian sample was found to be more sensitive to the outdoor temperature changes in AC mode (Fig. 9b). This trend was reversed in NV mode.

4.7.2. A comparison with ASHRAE's model

Fig. 9c shows the ASHRAE's adaptive comfort zones for Chennai (double hatched area) and Hyderabad (hatched area) along with the NBC's recommendations. Notably, the outdoor monthly mean temperature for Chennai and Hyderabad exceeds the upper applicability limit of 33 °C set forth in the ASHRAE's model. The outdoor temperature range in Hyderabad is wider than that of Chennai. As a result, we observed ASHRAE's adaptive comfort zone of Hyderabad (hatched area in Fig. 9c) enveloping Chennai on either side (double hatched area in Fig. 9c).

Taking a 'month' as a unit of analysis, we aggregated and superimposed the comfort data for all the floors in all the buildings in NV mode obtained in this survey on the ASHRAE's adaptive model. The aggregated data of comfort temperature were regressed with outdoor monthly mean temperature for comparison with the ASHRAE's adaptive model (slope: 0.31, comparable to that of the smaller SCATs dataset from Europe). We obtained the following relationship:

NV mode:

$$T_{\text{comf}} = 0.20T_{m-m} + 22.8 \quad (5)$$

(S.E. = 0.038, $r = 0.40$, $p < 0.001$, $N = 142$ aggregated observations).

While much of the data from this study falls within the 80% acceptability limits, a few points lie above these limits on the warmer side as well. It may be important to understand the discomfort during the temperature excursions and the adaptations that the subjects have undertaken.

4.7.3. Offset from comfort temperature (T_{diff})

When the indoor temperature differs from the comfort temperature, discomfort may be caused, with greater discomfort expected at higher deviations, or "comfort temperature drift". We estimated the comfort temperature for both NV and AC modes using the Eqs. (3) and (4) and obtained the comfort temperature drift from the coincident indoor temperature using the relationship [47]:

$$T_{\text{diff}} = T_g - T_{\text{comf}} \quad (6)$$

The likelihood of subjects voting at a particular point on the thermal sensation scale or lower for a given comfort temperature migration (T_{diff}) can be estimated using logistic regression analysis. The logistic regression lines of proportion voting on the TS scale and the corresponding T_{diff} values for various TS values are shown in Fig. 10.

This analysis was done on the raw data. To facilitate a comparison of the regression lines with the actual data, we ranked and divided the whole database (NV and AC mode) of T_{diff} values into 20 bins of approximately equal number of datasets in each bin. We then estimated the proportion of subjects voting at a particular TS scale point or less (for example, proportion voting at 2 or less, 1 or less etc.), in each of the T_{diff} bins. These points are superimposed over the regression lines. The actual proportion voting closely matched with the regression curves. A detailed description of this method can be found in Nicol and Humphreys [47].

These curves can be used to predict the percentage of population voting at or less than a given sensation vote at various degrees of offset from the comfort temperature. For example, about 80–90% subjects would vote below or at 'slightly warm' sensation when the indoor temperature drifts by about 1 K on the cooler side of the comfort temperature.

Occupants who are comfortable are generally expected to vote between -1 and $+1$ on the sensation scale. From the logistic curves shown above in Fig. 10, we can obtain the proportion voting

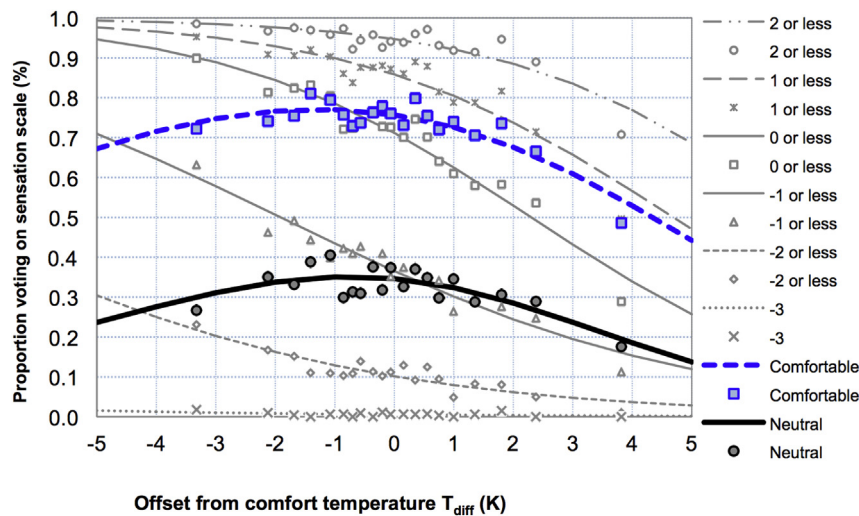


Fig. 10. Proportion of subjects voting comfortable (voting -1 to $+1$) (dashed line with open boxes) and neutral (voting 0) (solid line with closed circles) on the sensation scale varying with the offset from the comfort temperature. Also shown superimposed in the backdrop are the logistic regression curves showing the proportion of subjects voting at various points or lower on the thermal sensation scale. Actual proportions are shown superimposed in 20 bins of roughly equal size.

comfortable at various points of departure from the comfort temperature. The actual data overlapped the regression lines very closely, reinforcing the predictive power of the regression curves.

Using these lines we can predict that if the comfort temperature deviates by about +2 K, a little less than 70% subjects would be comfortable. Interestingly, when the comfort temperature oscillates by ± 1 K, proportion voting *neutral* would be between 32 and 35% in Indian offices as against 50–55% in European offices [47]. More importantly, 80% satisfaction levels would not be attained even when the indoor temperature drift is zero from the comfort temperature. On the contrary, from the SCAT surveys Nicol and Humphreys predicted that 80% would vote comfortable when the T_{diff} is within ± 2 K in European offices [47].

The reasons for lower satisfaction levels in Indian offices in our opinion are manifold; some of these possibly could be low indoor air velocities in AC mode (Fig. 4b and Table 6) and frequent daily power outages. The subjects yearned for higher air movement most of the time as explained in Indraganti et al. [10]. However, as observed in Fig. 9 much of the comfort data obtained from the field study remained above the NBC limits. Therefore, it is important to study the role of fans or the lack thereof.

4.8. Role of fans in providing thermal comfort

Fans offer an important adaptive opportunity for users at high indoor temperatures usually experienced in India. Nicol [49] shows that fan use increases the comfort temperature by about 2 K over a wide range of outdoor temperatures. Through field studies in north India and Iran, Nicol [50] noted air movement shifting the comfort temperature up by as much as 4 K, and in Roorkie, North India, Sharma and Ali [19] also identified similar a phenomenon. Rijal et al. [42], observed 81% fans on when the indoor temperature was 30 °C in Pakistan.

Zhang et al. [51] analysed the ASHRAE database and found comfort temperatures in the range of 25.5–28 °C with ceiling fans. They further observed that the use of additional personal fans pushed the upper boundary of comfort range to 30 °C. Other researchers have also found the use of fans significantly improving the indoor comfort [42]. In a climate chamber study, subjects accepted the thermal conditions at 28–32 °C under ceiling fans when the air velocity was above the ASHRAE limit of 0.8 m/s [52].

Similarly, Gong et al. [53] also identified an acceptable air velocity range from 0.3 up to 0.9 m/s under the experimental conditions. Others also noted thermal comfort being achieved

with personally controlled air movement, up to 30 °C and 60% RH, and acceptable perceived air quality being maintained up to 30 °C and 80% RH, without discomfort from humidity, air movement or eye-dryness [34]. They argue that 80% acceptable limit implicit in comfort standards could be extended to 30 °C and 60% RH [34].

4.8.1. Relationship between fan usage and the offset from comfort temperature

In all the field surveys we noted down the use (P_f) and availability of fans as nominal variables (*fan off*: 0; *fan on*: 1; 0: *Un-available*; 1: *available*). In order to predict the use of fans at various indoor conditions we analysed T_g with the binary variable '*fan usage*' (P_f) using logistic regression. This method is useful to predict the probability of a fan being in use (P_f) or not in use at a given indoor globe temperature (T_g). The logit can be calculated using the equation:

$$\text{Logit}(p) = \log(p/(1-p)) = bT + c \quad (7)$$

Where,

$$p = e^{(bT+c)} / (1 + e^{(bT+c)}) \quad (8)$$

Where p is the probability that the fan is in use, T is the temperature (in this case indoor globe temperature), b is the regression coefficient for T and c is the constant in the regression equation.

The logistic regression of P_f and T_g yielded the following equations with the following SE and Nagelkerke R^2 values:

NV mode:

$$\text{logit}(p) = 0.634T_g - 16.524 \quad (9)$$

($N = 1355$, SE = 0.06, $R^2 = 0.18$, $p < 0.001$).

AC mode:

$$\text{logit}(p) = 0.547T_g - 15.867 \quad (10)$$

($N = 1955$, SE = 0.04, $R^2 = 0.16$, $p < 0.001$).

These two equations are presented in Fig. 11. The actual proportion of fans on at 1 K interval bins of T_g is shown superimposed on the logistic curves. The logistic curves followed the actual proportion closely in both NV and AC modes.

These equations would be useful in simulation studies where it may be necessary to predict the use of fans as an adaptive mechanism. For example, at 29 °C nearly 87% fans in NV mode and 50% fans in AC mode would be on. Higher fan usage in NV mode in part can be explained by higher humidity in both the cities in NV mode (Fig. 4). In AC mode, humidity is mostly controlled by the HVAC system, and this has possibly allowed lower fan usage.

The use of the ceiling fan had a significant effect on the comfort temperatures. Use of fans increased the mean comfort temperature (T_{comf}) by 1.1 K and 0.6 K in NV and AC modes respectively. The improvement is statistically significant at 95% CI in both NV and AC modes.

The effect of fan use at low air speeds was found to be limited. However, at air speeds of about 1 m/s, the mean comfort temperature moved up by about 2.7 K in both NV and AC modes in this study. For example $T_{\text{comf-mean}}$ at air speeds of 1.0 m/s was as high as 30.1 °C and 28.6 °C in NV and AC modes respectively.

However, the effectiveness of fan as an adaptive mechanism hinges strongly on its availability and efficacy. Fans were available

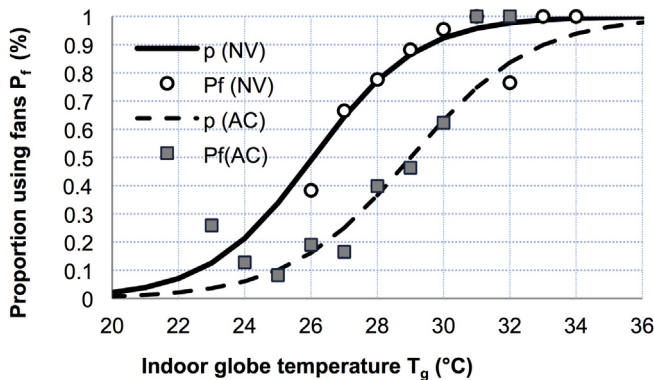


Fig. 11. Logistic regression of the proportion of fans in use as a function indoor globe temperature in NV (solid line and open circles) and AC modes (dashed line and closed boxes).

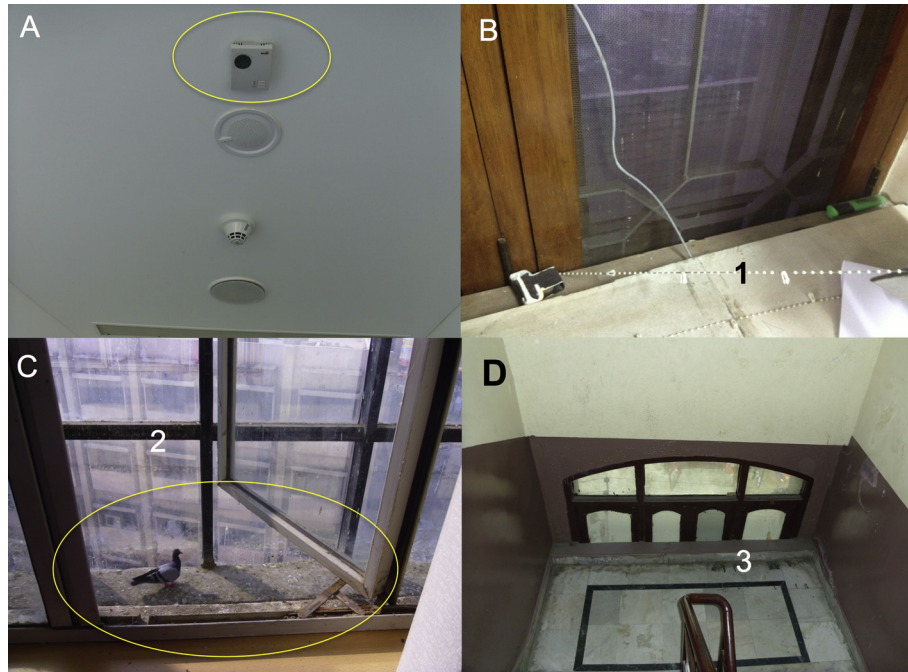


Fig. 12. Architectural and structural factors limiting thermal adaptation. (a) Thermostat controls permanently fitted to the ceiling; (b) Window operation blocked by wall-mounted tables (1); (c) Operable windows fixed immediately behind the structural glazing (2); (d) Structural floors (3) butting against the operable windows making them ineffective.

in most of the NV environments in both the cities. On the contrary, in AC mode, fans were found in 70% cases in Chennai and 11% in Hyderabad. Interestingly where fans were available, 20.6% and 68.5% fans in AC environments were ‘in use’ in Chennai and Hyderabad respectively during the 14 months of survey. The same figure was about 80% in NV mode for both the cities.

Coupled with non-availability, most of the offices suffered from inefficient fans. Their power output was 45–100 W, which sometimes also produced a lot of noise. This is quite in contrast to the fans found in use in the experiments of Zhai et al. [34] where the fans consumed typically 2–14 W. They noted that the fans consumed less than 10 W per person on an average for maintaining comfort, indicating that comfort can be delivered with energy-efficiency by providing air movement in warm-humid environments.

Therefore, efficiency and proliferation of fans must be improved to fully exploit the fan as an adaptive mechanism. There is a great opportunity for improvement on these fronts in India. In addition, it may also be necessary to take a look at a few other structural and architectural factors that affected thermal adaptation in offices in India.

4.9. Architectural and structural factors affecting thermal adaptation

We noticed several architectural and structural features adversely affecting thermal adaptation. Some of these are presented in Fig. 12. In some offices, we found the thermostat controls fitted to the ceiling. The occupants never adaptively operated these (Fig. 12a). Building or facility managers/service personnel controlled the thermostats in most of the offices. The interior design regulations were not strictly adhered to, in Hyderabad and Chennai. As a result, wall mounted tables were fitted against the windows blocking the window operation completely in some of the offices (Fig. 12b).

Anecdotal responses revealed some amount of user indifference and lethargy in operating the windows in general. In this context, it may be advantageous to consider installing window-signalling

systems that alert and motivate the users when to open/close a window. This may improve window operation as noted by Ackerly and Brager [54]. They reported that the occupants who discovered value in the signals were more likely to be more satisfied with their personal control.

Often times curtain glazing is fixed on to the external walls with windows, leaving just a narrow gap in between the glazing and window. The curtain glazing can thus obstruct the window operation (Fig. 12c). This small 150–300 mm wide space in between the curtain glazing and windows was found to be a nesting ground for pigeons or was dumped with trash in some other cases. This further restricted the window operation even during outages.

Sometimes intermediate floor slabs butted the operable windows making them totally ineffective, as shown in Fig. 12d. Indraganti et al. made similar observations, in a Tokyo study [36]. They noticed curtains/blinds obstructing the adaptive operation of vertically pivoted windows in the NV mode.

Constraints on occupants’ adaptive opportunities may lead to the failure of the adaptive model to predict thermal comfort responses in those circumstances. For instance, clothing adaptation was found to be restricted by office dress code in men and by some socio-cultural practices in women as noted in Section 4.2. Arens et al. [7] noted in a field study database that a clothing change from 0.5 clo to 1.0 clo corresponded to a 3 K air temperature change. Rijal et al. [37] also have noted even higher changes in comfort temperature with changes in clothing insulation. The *Cool Biz* and *Super Cool Biz* campaigns in Japan promoted relaxed work wear in offices in summer, thereby increasing the set temperatures to an advantage [36,55]. It may be advantageous for the building managers and users to reconsider their dress code norms and other clothing practices in light of the experiences from Japan and other parts of the globe.

5. Summary of findings

For fourteen months, high precision instruments and standard questionnaires were used to collect 6048 datasets from 2787

individuals in 28 NV and AC office buildings in two cities, Chennai and Hyderabad, having warm humid and composite climates. These two cities are in South India and face severe power shortages. Following are a summary of the findings.

1. Mean indoor globe temperatures in these buildings were around 28.8 °C in NV mode and 26.2 °C in AC mode and relative humidities generally between 45 and 48% RH in NV and AC modes. Indoor environments in Chennai were slightly warmer and had higher air speeds and humidities than those in Hyderabad. Indoor temperatures followed the outdoor temperature very closely in NV mode in both the cities. The 2005 Indian National Building Code specifies a comfort range of 21–26 °C. Most of the environments in NV mode and about half of them in AC mode were warmer than 26 °C. It would be pertinent to review this standard.
2. Mean clothing insulation was 0.7 clo, including 0.15 clo for upholstery insulation. It varied very slightly during the year and approximated closely to the *light clothing* (0.5–0.7) of ANSI/ASHRAE Std-55-2010. Metabolic rates were averaged to be 1.0 Met.
3. We developed an adaptive comfort model based on the outdoor running mean and indoor comfort temperatures for NV as $T_{\text{comf}} = 0.26 T_{\text{rm}} + 21.4$. This is in close agreement with the CEN standard (slope = 0.33 K⁻¹). Similar agreement was observed with the slope of the ASHRAE's adaptive model of 0.31 K⁻¹ as against 0.20 K⁻¹ obtained in this study.
4. Indoor comfort temperature in AC mode varied as $T_{\text{comf}} = 0.15 T_{\text{rm}} + 22.1$. Much similar relationship is included in the CIBSE guide (slope = 0.09 K⁻¹). Mean comfort temperature in NV mode was 28.0 °C and 26.4 °C in AC mode.
5. Air speeds of about 1.0 m/s pushed the comfort temperature up by about 2.7 K. Air speeds within the NV and AC modes were low, averaging at 0.17 m/s and 0.11 m/s respectively. In AC mode in Hyderabad an overwhelming majority had near still air conditions.
6. Fans were used adaptively to offset discomfort. Proliferation of fans was high in Chennai and was marginal in Hyderabad. Use of fans elevated the comfort temperature by about 0.7–1.8 K. The logistic regression predicted 87% and 50% fan usage in NV and AC modes when the indoor globe temperature would be 29 °C.

6. Conclusions and recommendations

The above results show that the subjects in Indian offices are comfortable at temperatures different from those suggested by the current Indian and international standards. We have noted occupants adapting in several ways to achieve comfort at elevated temperatures. An important and successful adaptation mechanism was the use of fans in both NV and AC offices. This suggests an opportunity to introduce higher air movement in Indian buildings to improve both energy and comfort performance. Given the scale of the scarcity of power, it may be the need of the hour.

India is climatologically and culturally very diverse. It may be necessary to undertake thermal comfort field studies in other climatic zones. This would provide a vast database, and would further the development of the adaptive standard for India. As Chappells and Shove [56] suggest, it may be prudent to promote a debate about the indoor environment and associated ways of life to avoid becoming locked into social and technical trajectories that are ultimately unsustainable, as comfort is a highly negotiable socio-cultural construct.

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